

Graph Neural Networks for Financial Fraud Detection: A Review

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Research Questions

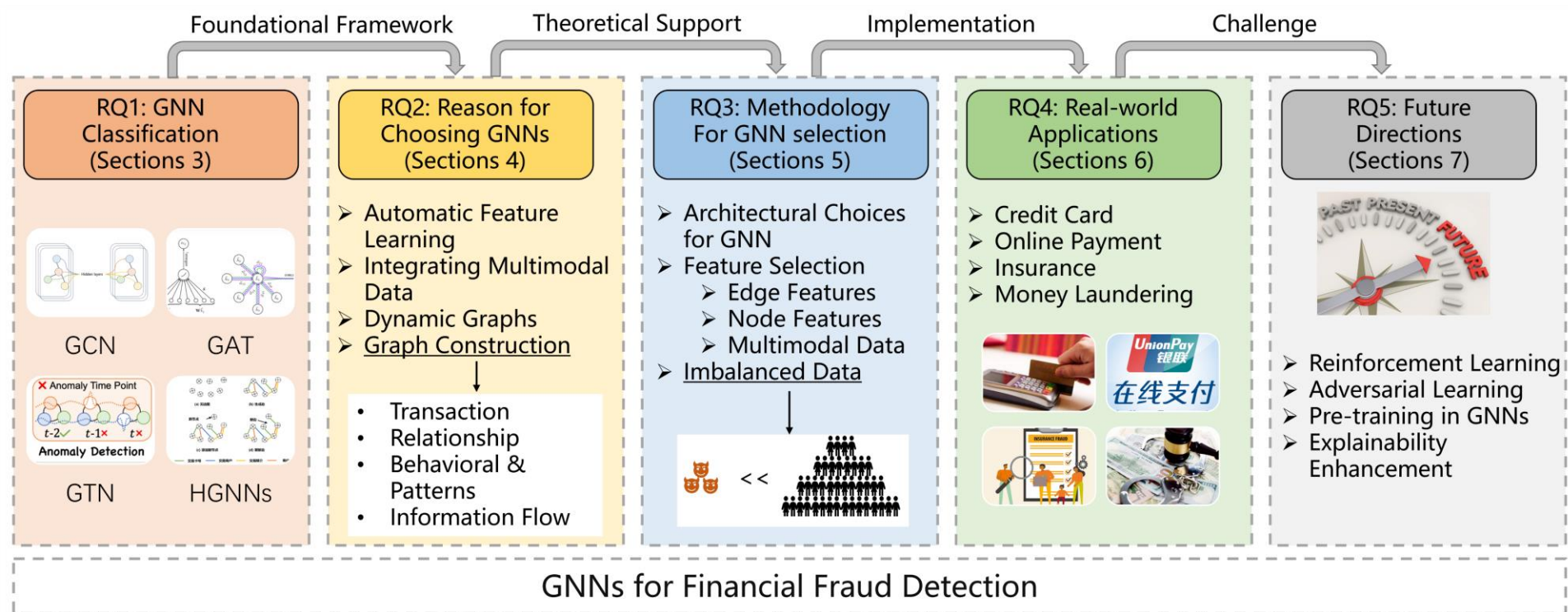
RQ1: How can various GNN methodologies in financial fraud detection be understood through a unified framework?

RQ2: Why use GNNs for financial fraud detection? What roles do they play?

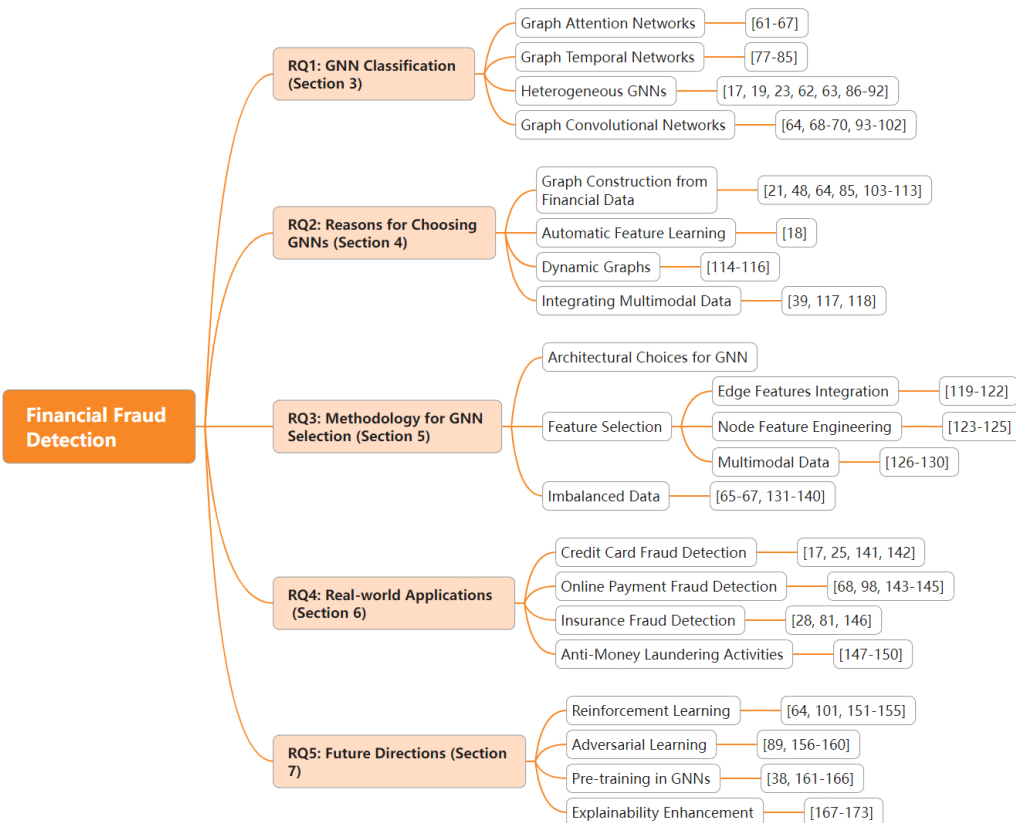
RQ3: How to design GNNs suitable for financial fraud detection?

RQ4: How are GNNs deployed in real-world financial fraud detection scenarios, and what are their impacts?

RQ5: What are the current challenges and future directions for GNNs in financial fraud detection?



Main Contributions



- ✓ **Unified Framework:** Proposed a framework to categorize and analyze GNN methods in financial fraud detection, enhancing understanding.
- ✓ **Suitability of GNNs:** Explored why GNNs are well-suited for tackling financial fraud detection.
- ✓ **Design Insights:** Provided insights into designing GNNs specifically for financial fraud detection.
- ✓ **Real-world Applications:** Highlighted practical examples of GNNs in action within the financial sector.
- ✓ **Future Directions:** Discussed current challenges and future opportunities in GNN research for fraud detection.

The general content of the paper